NILADRI BANERJEE

MSc in Data Science and AI, RKMVERI, Belur

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Github
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SUMMARY

My experience combines a strong foundation in machine learning and deep learning concepts with hands-on work in building end-to-end AI and data-driven systems. I am comfortable moving from understanding algorithms to implementing, training, and evaluating models in practical settings. Comfortable across the full lifecycle, I focus on building robust, privacy-preserved, and context-aware systems, systematically evaluating models using quantitative performance metrics to ensure efficiency and practical reliability.

EDUCATION

Belgharia High School, Belghaia
Secondary(Madhyamik), 2019 – 93%

BT Road Govt. Sponsd. High School
Higher Secondary, 2021 – 93.2%

Maulana Azad College, Kolkata
BSc (Honours) in Statistics, 2021–24
CGPA –6.58

Ramkrishna Mission Vivekananda Educational and Research Institute(RKMVERI)
MSc Data Science and Artificial Intelligence, 2025–27
CGPA – 6.09(till Sem 2)

SKILLS

- **Languages:** Python, R Programming, LaTeX
- **Frameworks:** OpenCV, Scikit-Learn, NumPy, Pandas, Streamlit, PyTorch, Matplotlib, Flask, Apache Hadoop, Apache Spark
- **Operating Systems:** Windows, Ubuntu Linux
- **Tools:** Git/GitHub, MS Office

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, Emerging Technologies(Large Language Models, Agentic AI, Blockchain), Cyber Security
- Advance Probability and Statistics, Linear Algebra, Time Series Analysis, Survival Ananlysis, Econometrics, Finance
- Distributed Computing, Data Structures and Algorithms, Database Management System

ACHIEVEMENTS

- GATE-DA 2026 – Qualified
- CUET-PG 2025 (Statistics) – Qualified
- IIT JAM 2025 (Statistics) – Qualified
- NMMS SCHOLAR 2016-17

Positions of Responsibility

- Designer Volunteer – Perceptron 2025-26 (The Annual Fest, RKMVERI)
- Class Representative – RKMVERI, Belur (2025-27 batch)
- Convenor – Perceptron 2026-27 (The Annual Fest, RKMVERI)

PROJECTS

Agentic RAG FinOps Auditor with Local LLMs
RKMVERI June 2026 – July 2026

Architected an enterprise financial compliance engine using LangGraph, LangChain, and ChromaDB running locally. Integrated Ollama (Llama-3) and MiniLM vector embeddings to convert unstructured policy PDFs into structured, context-grounded audit execution workflows without external API dependencies. Enabled real-time anomaly detection for SaaS procurements and non-compliant weekend operational charges, ensuring complete enterprise data privacy.

Automated Invoice Matching & Anomaly Detection
RKMVERI Feb 2026 – Apr 2026

Engineered a multimodal pipeline integrating CNN-based OCR and a fine-tuned Legal-BERT Transformer. Developed an unsupervised Deep Autoencoder forensic layer to flag high-risk anomalies via reconstruction error analysis.

RL-Based Adaptive GUI & Web Automation Agent
RKMVERI Feb 2026 – Apr 2026

Designed and implemented an end-to-end Reinforcement Learning framework progressing from grid-world simulation to real browser automation. Trained Q-Learning and Deep Q-Network (DQN) agents to learn optimal interaction policies in dynamic and stochastic environments. Applied Transfer Learning to accelerate adaptation across changing GUI layouts and integrated Selenium-driven browser automation for autonomous login workflow execution.

Distributed Fraud Detection using Apache Spark & H2O.ai
RKMVERI Feb 2026 – Apr 2026

EDeveloped a distributed credit card fraud detection system using Apache Spark, PySpark, H2O.ai DRF, and Sparkling Water. Built scalable data preprocessing, class-balanced model training, validation-based threshold optimization, distributed prediction, and evaluation using AUC, PR-AUC, F1 Score, Recall, and Balanced Accuracy.

Warehouse Robot Navigation Using Deep RL
RKMVERI Oct 2025 – Dec 2025

Modeled a warehouse pickup-and-delivery task as a Markov Decision Process (MDP) and built a custom Gymnasium environment with directional wall constraints. Trained a Deep Q-Network (DQN) agent in PyTorch using custom reward structures and an ϵ-greedy schedule. Achieved reliable policy convergence with a significant task success rate, outperforming baseline navigation methods

Image Classification using Machine Learning
RKMVERI Sep 2025 – Nov 2025

Built an end-to-end classical ML pipeline for Hollywood celebrity facial classification using 64×64 grayscale images and a 4096-dimensional hybrid representation combining raw pixel and Haar wavelet features. Applied training-only standardization and PCA-based dimensionality reduction before evaluating RBF-SVM, Random Forest, and Logistic Regression classifiers. Additionally, conducted controlled feature ablation experiments comparing raw-pixel, wavelet, and hybrid representations.